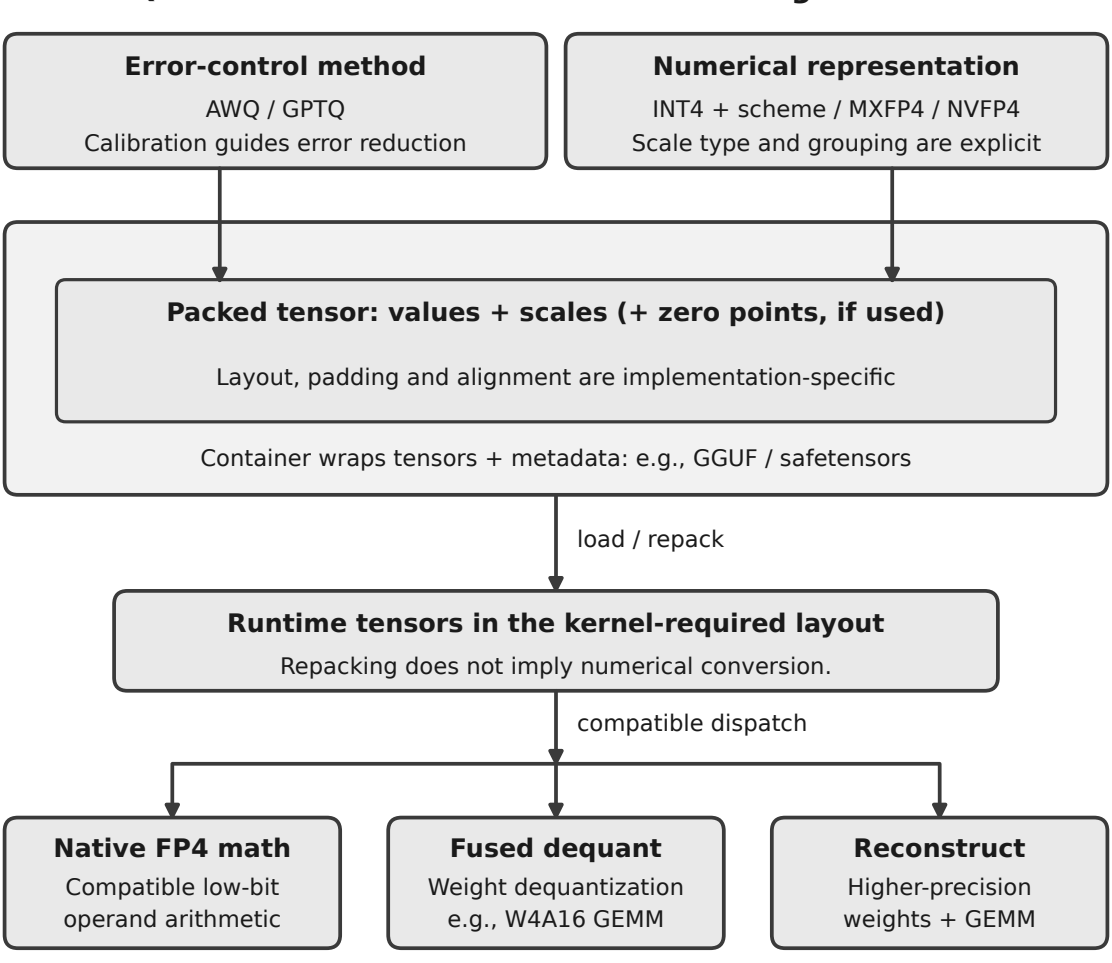
Quantization choices are not execution guarantees
Error-control method
AWQ / GPTQ
Calibration guides error reduction
Numerical representation
INT4 + scheme / MXFP4 / NVFP4
Scale type and grouping are explicit
Packed tensor: values + scales (+ zero points, if used)
Layout, padding and alignment are implementation-specific
Container wraps tensors + metadata: e.g., GGUF / safetensors
load / repack
Runtime tensors in the kernel-required layout
Repacking does not imply numerical conversion.
compatible dispatch
Native FP4 math
Compatible low-bit
operand arithmetic
Fused dequant
Weight dequantization
e.g., W4A16 GEMM
Reconstruct
Higher-precision
weights + GEMM
Only supported combinations can load and execute.
No compatible kernel: reject; fallback is not guaranteed.